

| PRODUCT    | MEMBER #   | MIL | GAGE | COATING  |
|------------|------------|-----|------|----------|
| Smart Stud | 400S137-97 | 97  | 12   | G60, G90 |

400S137-97 Product Submittal

PHYSICAL PROPERTIES

|                        |          |
|------------------------|----------|
| Web Depth              | 4"       |
| Flange                 | 1.375"   |
| Return (lip)           | 0.375"   |
| Approx. Weight (lb/ft) | 2.3375   |
| Area (in^2)            | 0.6859   |
| Design Thickness       | 0.1017   |
| Minimum Thickness      | 0.0966   |
| Yield Strength         | 50 KSI   |
| Coating                | G60, G90 |

GROSS PROPERTIES

|                                                     |        |
|-----------------------------------------------------|--------|
| Gross section moment of inertia about X-X axis (Ix) | 1.5580 |
| Section modulus about X-X axis (Sx)                 | 0.7790 |
| Radius of gyration about X-X axis (Rx)              | 1.507  |
| Gross section moment of inertia about Y-Y axis (Iy) | 0.1420 |
| Radius of gyration about Y-Y axis (Ry)              | 0.455  |

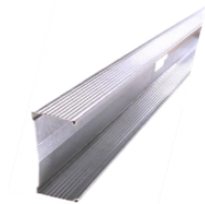
EFFECTIVE / NET PROPERTIES

|                                          |        |
|------------------------------------------|--------|
| Effective moment of inertia X-axis (Ixe) | 1.5627 |
| Effective section modulus X-axis (Sxe)   | 0.7832 |
| Allowable moment (Mal)                   | 23.450 |
| Allowable shear (Vay)                    | 6.658  |

TORSIONAL PROPERTIES

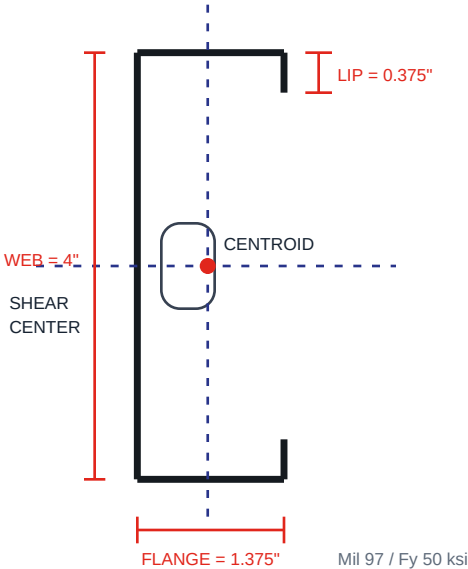
|                                        |        |
|----------------------------------------|--------|
| St. Venant torsional constant (Jx1000) | 2.3646 |
| Torsional warping constant (Cw)        | 0.4858 |
| Polar radius of gyration (Ro)          | 1.806  |

PRODUCT VIEW



CAD SECTION

schematic dimensions



NOTES

Code Approvals & Performance Standards

- AISI S100-16 (2020) w/S2-20 North American Specification for the Design of Cold-Formed Steel Structural Members.
- AISI S240-20 North American Standard for Cold-Formed Steel Structural Framing.
  - o Compliant to ASTM C955; IBC replaced ASTM C955 with AISI S200 in IBC 2015 and AISI S240 in IBC 2018.
  - o Section A3 Material - Chemical and mechanical requirements (referencing ASTM A1003/A1003M).
  - o Section A4 Corrosion Protection (referencing ASTM A653/A653M).
  - o Section A5 Products - Thickness, shapes, tolerances, identification.
  - o Section C Installation (referencing ASTM C1007).
- AISI S202-20 Code of Standard Practice for Cold-Formed Steel Structural Framing.
  - o Section F3 Delivery, Handling, and Storage of Materials.
- IBC 2024 International Building Code.
- ICC-ES ESR-5837 Structural Studs and Track.
  - o ESR-5837 Catalog Refined Metal Technical Design Guide (10/1/2025).
- SDS for ASTM A1003 Steel Framing Products for interior framing, exterior framing, and clips/accessories.
- The technical content of this literature is effective 08/26/25 and supersedes all previous information.

DISCLAIMER

All data, details, and specifications included herein are intended as a general guide for using Refined Metal products. These products should not be used in design or construction without evaluation by a qualified engineer or architect to determine suitability for a specific use. Refined Metal assumes no liability for failure resulting from use or misapplication of computations, details, or specifications contained herein. Refined Metal assumes no liability for damages resulting from improper application or installation of these products.

DERIVED SMART STUD SPAN TABLES

Preliminary single-span estimates in feet. Verify with manufacturer span tables or sealed engineering before use.

|            | 15 psf dead load, 40 psf live load |       | 15 psf dead load, 60 psf live load |       | 15 psf dead load, 125 psf live load |       |
|------------|------------------------------------|-------|------------------------------------|-------|-------------------------------------|-------|
| Spacing OC | L/360                              | L/480 | L/360                              | L/480 | L/360                               | L/480 |
| 12         | 10.6                               | 9.6   | 9.6                                | 8.7   | 7.8                                 | 7.1   |
| 16         | 9.7                                | 8.8   | 8.7                                | 7.9   | 7.1                                 | 6.5   |
| 24         | 8.5                                | 7.7   | 7.7                                | 7.0   | 6.2                                 | 5.7   |

|            | 40 psf dead load, 40 psf live load |       | 40 psf dead load, 60 psf live load |       | 40 psf dead load, 125 psf live load |       |
|------------|------------------------------------|-------|------------------------------------|-------|-------------------------------------|-------|
| Spacing OC | L/360                              | L/480 | L/360                              | L/480 | L/360                               | L/480 |
| 12         | 9.4                                | 8.5   | 8.7                                | 7.9   | 7.4                                 | 6.7   |
| 16         | 8.6                                | 7.8   | 8.0                                | 7.2   | 6.7                                 | 6.1   |
| 24         | 7.5                                | 6.8   | 7.0                                | 6.3   | 5.9                                 | 5.4   |

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